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 Studierichting: Informatica
 Oefeningenreeks 7C nr. 2.2

Opgave: Bereken alle tweede partiële afgeleiden:

$$f : \mathbb{R}^2 \rightarrow \mathbb{R} : (x, y) \mapsto \sqrt{4x^2 + y^2}$$

$$\frac{\partial f}{\partial x} = \frac{8x}{2\sqrt{4x^2 + y^2}} = \frac{4x}{\sqrt{4x^2 + y^2}}$$

$$\frac{\partial f}{\partial y} = \frac{2y}{2\sqrt{4x^2 + y^2}} = \frac{y}{\sqrt{4x^2 + y^2}}$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{4\sqrt{4x^2 + y^2} - 4x \cdot \frac{4x}{\sqrt{4x^2 + y^2}}}{4x^2 + y^2} = \frac{4(4x^2 + y^2) - 16x^2}{(4x^2 + y^2)\sqrt{4x^2 + y^2}} = \frac{4y^2}{\sqrt{(4x^2 + y^2)^3}}$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{\sqrt{4x^2 + y^2} - y \cdot \frac{y}{\sqrt{4x^2 + y^2}}}{4x^2 + y^2} = \frac{(4x^2 + y^2) - y^2}{(4x^2 + y^2)\sqrt{4x^2 + y^2}} = \frac{4x^2}{\sqrt{(4x^2 + y^2)^3}}$$

$$\frac{\partial^2 f}{\partial x \partial y} = 4x \left(-\frac{1}{2} (4x^2 + y^2)^{-\frac{3}{2}} \cdot 2y \right) = \frac{-4xy}{\sqrt{(4x^2 + y^2)^3}}$$

$$\frac{\partial^2 f}{\partial y \partial x} = y \left(-\frac{1}{2} (4x^2 + y^2)^{-\frac{3}{2}} \cdot 8x \right) = \frac{-4xy}{\sqrt{(4x^2 + y^2)^3}}$$

$$D^2 f = \begin{pmatrix} \frac{4y^2}{\sqrt{(4x^2 + y^2)^3}} & \frac{-4xy}{\sqrt{(4x^2 + y^2)^3}} \\ \frac{-4xy}{\sqrt{(4x^2 + y^2)^3}} & \frac{4x^2}{\sqrt{(4x^2 + y^2)^3}} \end{pmatrix}$$

References